

DAY 7 — EMBEDDINGS + CHROMADB + SEMANTIC RETRIEVAL

GOAL

Move from:

clean tables 

to:

semantic searchable engineering knowledge base 

DAY 7 ARCHITECTURE

PDF

↓

Tables

↓

Clean tables

↓

Chunking

↓

Embeddings

↓

ChromaDB

↓

Semantic Search

WHAT DAY 7 ADDS

 embedding generation

 ChromaDB vector store

 chunk creation

 semantic retrieval

 similarity search

✓ engineering query search

✓ RAG foundation

✓ TEMPLATE STRUCTURE

app/

└─ embeddings/

| └─ embedder.py ✓

| └─ chunker.py ✓

| └─ vectordb.py ✓

|

└─ query/

| └─ semantic_search.py ✓

|

└─ ingestion/

| └─ pipeline.py ✓ MODIFY

✓ STEP 1 — INSTALL

pip install chromadb sentence-transformers

✓ STEP 2 — CHUNKER

📄 **app/embeddings/chunker.py**

```
def dataframe_to_chunks(table):
```

```
    chunks = []
```

```
    df = table["df"]
```

```
    page = table["page"]
```

```
table_id = table["table_id"]
```

```
for idx, row in df.iterrows():
```

```
    values = [  
        str(v)  
        for v in row.values  
    ]
```

```
    text = " | ".join(values)
```

```
    chunks.append({  
        "id": f"{table_id}row{idx}",  
        "text": text,  
        "metadata": {  
            "page": page,  
            "table_id": table_id  
        }  
    })
```

```
return chunks
```

✅ STEP 3 — EMBEDDER

 **app/embeddings/embedder.py**

```
from sentence_transformers import (  
    SentenceTransformer  
)
```

```
model = SentenceTransformer(  
    "all-MiniLM-L6-v2"  
)  
  
def create_embedding(text):  
  
    return model.encode(text).tolist()
```

✅ STEP 4 — CHROMADB

 **app/embeddings/vectordb.py**

```
import chromadb  
  
client = chromadb.PersistentClient(  
    path="./chroma_db"  
)  
  
collection = client.get_or_create_collection(  
    name="dvp_tables"  
)  
  
def store_chunks(chunks):  
  
    ids = []  
    docs = []  
    metas = []  
  
    for chunk in chunks:  
  
        ids.append(chunk["id"])
```

```
docs.append(chunk["text"])
```

```
metas.append(chunk["metadata"])
```

```
collection.add(  
    ids=ids,  
    documents=docs,  
    metadatas=metas  
)
```

✅ STEP 5 — SEMANTIC SEARCH

 **app/query/semantic_search.py**

```
from app.embeddings.vectordb import (  
    collection  
)
```

```
def semantic_search(query, top_k=5):
```

```
    results = collection.query(  
        query_texts=[query],  
        n_results=top_k  
    )
```

```
    return results
```

✅ STEP 6 — UPDATE PIPELINE

 **app/ingestion/pipeline.py**

```
from app.parser.page_builder import (  
    build_page_view  
)
```

```
from app.parser.pdfplumber_table import (  
    extract_tables_pdf  
)
```

```
from app.parser.table_normalizer import (  
    normalize_tables  
)
```

```
from app.parser.quality_scorer import (  
    filter_useful_tables  
)
```

```
from app.embeddings.chunker import (  
    dataframe_to_chunks  
)
```

```
from app.embeddings.vectordb import (  
    store_chunks  
)
```

```
def process_pdf(file_path):
```

```
    pages = build_page_view(file_path)
```

```
    raw_tables = extract_tables_pdf(
```

```
    file_path
)

normalized_tables = normalize_tables(
    raw_tables
)

useful_tables = filter_useful_tables(
    normalized_tables
)

all_chunks = []

for table in useful_tables:

    chunks = dataframe_to_chunks(
        table
    )

    all_chunks.extend(chunks)

store_chunks(all_chunks)

return {
    "pages": pages,
    "tables": useful_tables,
    "chunks": all_chunks
}
```

✅ STEP 7 — TEST INGESTION

 **tests/test_ingestion.py**

```
from app.ingestion.pipeline import process_pdf
```

```
result = process_pdf(
```

```
    r"D:\new_train\training\Pdf_praser_llm\data\0183-VP-TDM-06-08196-C006-1339-BRAKE THERMAL  
    MAPPING CITY CONDITION Vehicle Dynamics.pdf"
```

```
)
```

```
print("\n✅ TOTAL CHUNKS:")
```

```
print(len(result["chunks"]))
```

✅ STEP 8 — TEST SEARCH

 **tests/test_search.py**

```
from app.query.semantic_search import (
```

```
    semantic_search
```

```
)
```

```
results = semantic_search(
```

```
    "maximum front pad temperature"
```

```
)
```

```
print(results)
```

✅ EXPECTED RESULT

Front pad peak temperature | Maximum | 163 | 179 | 188

retrieved semantically 

✔ WHAT DAY 7 ACHIEVES

- ✔ engineering semantic search
 - ✔ vector database
 - ✔ embeddings
 - ✔ retrieval layer
 - ✔ RAG foundation
 - ✔ scalable architecture
-

✔ YOUR SYSTEM NOW

PDF

↓

Extraction

↓

Normalization

↓

Filtering

↓

Chunking

↓

Embeddings

↓

ChromaDB

↓

Semantic Search ✔

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✔ YOU ARE NOW HERE

✔ DAY 7 COMPLETE

Next:

DAY 8

- Hybrid Search
- BM25 + Vector
- Metadata Filtering
- Better engineering retrieval